Математика

№4

1) $\cos 315^\circ + \sin 210^\circ + tg\ 420^\circ$

$\cos 315^\circ = \cos(360^\circ - 45^\circ) = \cos 45^\circ = \frac{\sqrt{2}}{2}$

$\sin 210^\circ = \sin(180^\circ + 30^\circ) = -\sin 30^\circ = -\frac{1}{2}$

$tg\ 420^\circ = tg(360^\circ + 60^\circ) = tg\ 60^\circ = \sqrt{3}$

Ответ: $\frac{\sqrt{2}-1}{2} + \sqrt{3}$

2) $\sin \frac{13\pi}{6} - \cos \frac{11\pi}{6} + ctg \frac{11\pi}{4}$

$\sin \frac{13\pi}{6} = \sin(2\pi + \frac{\pi}{6}) = \sin \frac{\pi}{6} = \frac{1}{2}$

$\cos \frac{11\pi}{6} = \cos(2\pi - \frac{\pi}{6}) = \cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$

$ctg \frac{11\pi}{4} = ctg(2\pi + \frac{3\pi}{4}) = ctg \frac{3\pi}{4} = -1$

Ответ: $-\frac{1}{2} - \frac{\sqrt{3}}{2}$

№2

Дано: $\sin a = 0,2$ и $\frac{\pi}{2} \le a < \pi$.

1. Поскольку

$\cos a = -\sqrt{1 - \sin^2 a} = -\sqrt{1 - 0,2^2} = -\sqrt{1 - 0,04} = -\sqrt{0,96}$

$= -0,4\sqrt{6}$

2. Найдем:

$$tg\ a = \frac{\sin a}{\cos a} = \frac{0,2}{-\sqrt{0,96}} = -\frac{1}{\sqrt{24}} = -\frac{1}{2\sqrt{6}} = -\frac{\sqrt{6}}{12}$$

3. Наконец:

$$ctg\ a = \frac{1}{tg\ a} = -\sqrt{24} = -2\sqrt{6}$$

Ответ: $\cos a = -0,4\sqrt{6}$, $tg\ a = -\frac{\sqrt{6}}{12}$, $ctg\ a = -2\sqrt{6}$